

Results and Discussion

The performance of the Nginx VNF was evaluated in both cloud-like and edge-like Kubernetes environments using ApacheBench and iperf3. The results are summarised below.

Table 1: Application Performance (ApacheBench)

Metric	Cloud Environment	Edge Environment
Requests/sec	5382.83 req/sec	5985.48 req/sec
Latency	9.289 ms	8.354 ms
Failed Requests	0	0

```
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Completed 700 requests
Completed 800 requests
Completed 900 requests
Completed 1000 requests
Finished 1000 requests

Server Software:      nginx/1.29.6
Server Hostname:      127.0.0.1
Server Port:          61400

Document Path:        /
Document Length:      896 bytes

Concurrency Level:    50
Time taken for tests:  0.167 seconds
Complete requests:    1000
Failed requests:       0
Total transferred:    1129000 bytes
HTML transferred:     896000 bytes
Requests per second:  5985.48 [#/sec] (mean)
Time per request:     8.354 [ms] (mean)
Time per request:     0.167 [ms] (mean, across all concurrent requests)
Transfer rate:        6599.22 [Kbytes/sec] received

Connection Times (ms)
  min  mean[+/-sd] median   max
Connect:  0   0.2   0       5
Processing:  8   1.7   8      14
Waiting:    7   1.8   7      13
Total:      8   1.7   8      14

Percentage of the requests served within a certain time (ms)
 50%    8
 66%    8
 75%    9
 80%    9
 90%   10
 95%   12
 98%   13
 99%   14
100%   14 (longest request)
```

```
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Completed 700 requests
Completed 800 requests
Completed 900 requests
Completed 1000 requests
Finished 1000 requests

Server Software:      nginx/1.29.6
Server Hostname:      127.0.0.1
Server Port:          63951

Document Path:        /
Document Length:      896 bytes

Concurrency Level:    50
Time taken for tests:  0.166 seconds
Complete requests:    1000
Failed requests:       0
Total transferred:    1129000 bytes
HTML transferred:     896000 bytes
Requests per second:  5382.83 [#/sec] (mean)
Time per request:     9.289 [ms] (mean)
Time per request:     0.166 [ms] (mean, across all concurrent requests)
Transfer rate:        5934.78 [Kbytes/sec] received

Connection Times (ms)
  min  mean[+/-sd] median   max
Connect:  0   0.4   0      13
Processing:  8   2.1   8      18
Waiting:    7   2.0   7      16
Total:      8   2.1   8      21

Percentage of the requests served within a certain time (ms)
 50%    8
 66%    9
 75%    9
 80%   10
 90%   11
 95%   11
 98%   13
 99%   15
100%   21 (longest request)
```

Table 2: Network Performance (iperf3)

Metric	Cloud	Edge
Bandwidth	852 Mbits/sec	887 Mbits/sec

```
PS C:\Users\orste\OneDrive - University of Leeds\Desktop\PPT Of All\Cloud Co
mputing\iperf> .\iperf3.exe -c 127.0.0.1
Connecting to host 127.0.0.1, port 5201
[ 4] local 127.0.0.1 port 64918 connected to 127.0.0.1 port 5201
[ ID] Interval      Transfer    Bandwidth
[ 4] 0.00-1.01 sec  92.5 MBytes 767 Mbits/sec
[ 4] 1.01-2.01 sec  92.0 MBytes 776 Mbits/sec
[ 4] 2.01-3.00 sec  111 MBytes 934 Mbits/sec
[ 4] 3.00-4.01 sec  110 MBytes 921 Mbits/sec
[ 4] 4.01-5.01 sec  98.2 MBytes 826 Mbits/sec
[ 4] 5.01-6.00 sec  81.5 MBytes 687 Mbits/sec
[ 4] 6.00-7.00 sec  108 MBytes 906 Mbits/sec
[ 4] 7.00-8.01 sec  118 MBytes 977 Mbits/sec
[ 4] 8.01-9.00 sec  92.2 MBytes 781 Mbits/sec
[ 4] 9.00-10.01 sec 113 MBytes 939 Mbits/sec
-----
[ ID] Interval      Transfer    Bandwidth
[ 4] 0.00-10.01 sec 1016 MBytes 852 Mbits/sec                               sender
[ 4] 0.00-10.01 sec 1016 MBytes 852 Mbits/sec                               receiver

iperf Done.
PS C:\Users\orste\OneDrive - University of Leeds\Desktop\PPT Of All\Cloud Co
mputing\iperf> kubectl config current-context minikube
```

```
mputing\iperf> .\iperf3.exe -c 127.0.0.1
Connecting to host 127.0.0.1, port 5201
[ 4] local 127.0.0.1 port 58050 connected to 127.0.0.1 port 5201
[ ID] Interval      Transfer    Bandwidth
[ 4] 0.00-1.01 sec  86.9 MBytes 719 Mbits/sec
[ 4] 1.01-2.00 sec  86.2 MBytes 732 Mbits/sec
[ 4] 2.00-3.00 sec  71.8 MBytes 601 Mbits/sec
[ 4] 3.00-4.00 sec  183 MBytes 1.54 Gbits/sec
[ 4] 4.00-5.01 sec  73.2 MBytes 611 Mbits/sec
[ 4] 5.01-6.01 sec  95.6 MBytes 796 Mbits/sec
[ 4] 6.01-7.01 sec  87.4 MBytes 734 Mbits/sec
[ 4] 7.01-8.01 sec  68.9 MBytes 579 Mbits/sec
[ 4] 8.01-9.00 sec  96.0 MBytes 813 Mbits/sec
[ 4] 9.00-10.00 sec 209 MBytes 1.75 Gbits/sec
-----
[ ID] Interval      Transfer    Bandwidth
[ 4] 0.00-10.00 sec 1.03 GBytes 887 Mbits/sec                               sender
[ 4] 0.00-10.00 sec 1.03 GBytes 887 Mbits/sec                               receiver

iperf Done.
PS C:\Users\orste\OneDrive - University of Leeds\Desktop\PPT Of All\Cloud Co
mputing\iperf> kubectl config current-context edge
```

Table 3: Resource Utilisation

Metric	Cloud	Edge
Memory Usage	6 Mi	5 Mi
CPU Usage	Minimal	Minimal

```
PS C:\xampp\apache\bin> kubectl top pods
NAME                                CPU(cores)  MEMORY(bytes)
nginx-56c45fd5ff-gc7rz             0m          6Mi
```

```
PS C:\xampp\apache\bin> kubectl top pods
NAME                                CPU(cores)  MEMORY(bytes)
nginx-56c45fd5ff-24v9c             0m          5Mi
```

Discussion of Results

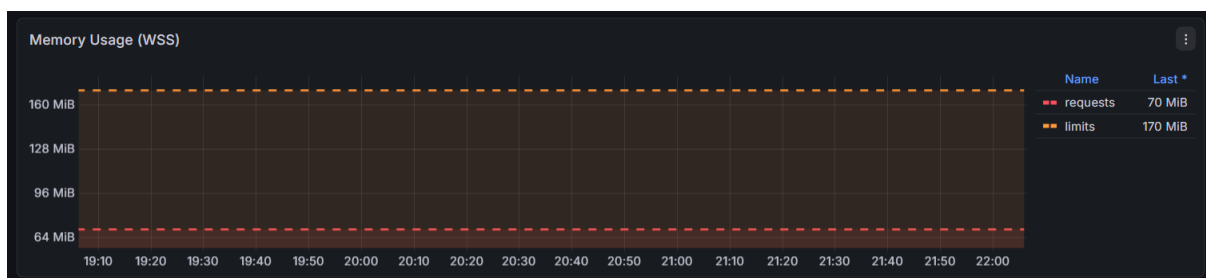
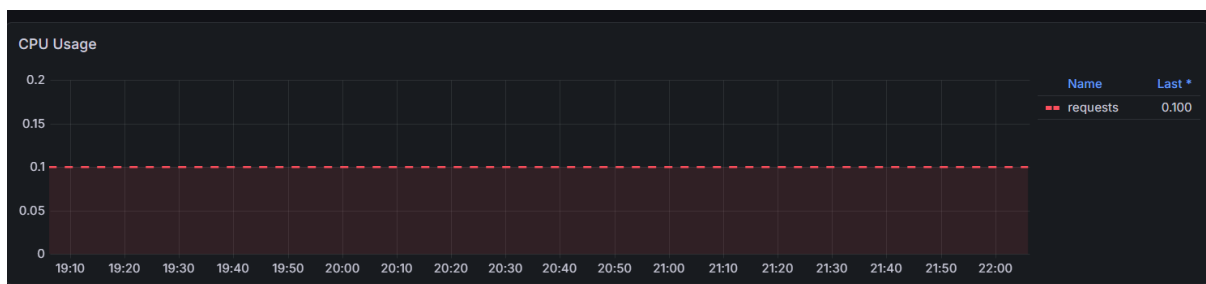
It can be observed from the result that both environments maintained effective performance throughout the entire workload testing period. The edge environment showed slightly higher throughput and lower latency compared to the cloud environment. However, it is minimal and doesn't have a significant effect.

This behaviour can be explained by the fact that both environments were executed on the same physical machine using Minikube. As a result, network conditions, hardware resources, and system overhead were nearly identical. This eliminates real-world variations such as network delay, distributed infrastructure, and hardware heterogeneity.

(Note: I have tried using microk8s for edge and establishing a VM on azure but both didn't work out as I have mentioned in my above answers)

The iperf3 results confirm this finding because both systems achieved stable high bandwidth performance with only minor bandwidth differences. The two systems utilized resources at minimal levels which confirmed Nginx operates as a lightweight VNF while producing minimal processing demands.

From a visualization perspective, Grafana dashboards showed only minor fluctuations in CPU and memory usage during load testing. This is expected given the lightweight nature of the workload and the local deployment environment.



When contrasted with existing literature, edge computing is generally expected to provide lower latency due to proximity to end users, while cloud environments offer higher scalability and resource availability. However, in this experiment, such differences were not prominently observed due to the controlled and localised setup. This highlights a limitation of using Minikube for performance evaluation, as it does not fully replicate real-world distributed systems.

Conclusion

Overall, the results demonstrate that Kubernetes can efficiently run VNFs in both cloud and edge-like environments. While performance differences were minimal in this setup, the experiment validates the feasibility of deploying lightweight network functions at the edge.